A COMPACTIFYABLE SCHEME MAPPING TO A SEPARATED FINITE TYPE SCHEME HAS A COMPACTIFICATION OVER THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0A9Z (sources/stacks-flat/pinned-original-source.tex line 9387).

Allowed standard prerequisites (editorial).

- Scheme theoretic image of a partial section and closedness of the semi-diagonal. Morphisms, Lemma labelled morphisms-lemma-scheme-theoretic-image-of-partial-section (Stacks tag 0CNG) invoked at proof lines 9403-9405, and Schemes, Lemma labelled schemes-lemma-semi-diagonal (Stacks tag 01KS) invoked at proof lines 9401-9402. Supply that X -> Xbar' is an open immersion once (1, f) : X -> X x_S Y is a closed immersion and the partial section s = (j, f) is used.
- Compactifications and compactifyable schemes over a quasi-compact quasi-separated base. Same chapter, flat.tex, the definition of compactifyable schemes and of a compactification over S used in the statement. Supplies the starting compactification j : X -> Xbar and the properness of Xbar -> S that is base changed to Y.

Source reference. Original lemma 01KS: sources/stacks-chow/tags.txt, source lines 2061–2061. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9402 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0CNG: sources/stacks-chow/tags.txt, source lines 15891–15891. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 9404 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a quasi-compact and quasi-separated scheme. Let $f : X \to Y$ be a morphism of schemes over S with Y separated and of finite type over S and X compactifyable over S. Then X has a compactification over Y.*

Proof. Let $j : X \to \overline{X}$ be a compactification of X over S. Then we let $\overline{X}'$ be the scheme theoretic image of $(j, f) : X \to \overline{X} \times_S Y$. The morphism $\overline{X}' \to Y$ is proper because $\overline{X} \times_S Y \to Y$ is proper as a base change of $\overline{X} \to S$. On the other hand, since Y is separated over S, the morphism $(1, f) : X \to X \times_S Y$ is a closed immersion (Schemes, Lemma 01KS) and hence $X \to \overline{X}'$ is an open immersion by Morphisms, Lemma 0CNG applied to the "partial section" $s = (j, f)$ to the projection $\overline{X} \times_S Y \to \overline{X}$. □